

Zhijun Zeng

Tsinghua University Yau Mathematical Sciences Center (YMSC)

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Education

Ph.D. Computational Mathematics, <i>Tsinghua University</i> . Advisors: Yi Zhu / Zuoqiang Shi	2022–Present
B.S. Mathematics and Applied Mathematics, <i>Shanghai University of Finance and Economics</i>	2018–2022
Visiting Scholar , <i>Cornell University</i> , Ithaca, USA	Fall 2024

Work Experience

AI4S Algorithm Intern , Operator learning for accelerated ultrasound computed tomography <i>DP Technology</i>	2023–2025
AI4S Algorithm Intern , Operator-learning-based flow control <i>Microsoft Research Asia (MSRA)</i>	2022–2023
Algorithm Intern , Phase-field simulation and algorithm design <i>Beijing Computational Science Research Center (CSRC)</i>	2021

Honors & Funding

YMSC Full Fellowship for Ph.D. Students , Tsinghua University	2025–2026
Tsinghua Ph.D. Short-term Visit Fund , Visiting Scholar at Cornell University	2024–2025
National Second Prize , China Undergraduate Mathematical Contest in Modeling (CUMCM)	2021
National Scholarship	2021

Research Interests

- Computational inverse problems; ultrasound imaging; nonlinear filtering; knowledge discovery and system identification; diffusion language models; science agents.

Publications

Computational Imaging

1. Zeng Z, Zheng Y, Su C, *et al.* Ultrasound Tomography of Musculoskeletal Tissues with Generative Neural Physics. *arXiv preprint* arXiv:2508.12226, 2025. (Accepted by *ICCP 2026*; Recommended for *IEEE TPAMI*)
2. Zeng Z, Zheng Y, Hu H, *et al.* OpenBreastUS: Benchmarking Neural Operators for Wave Imaging Using Breast Ultrasound Computed Tomography. *arXiv preprint* arXiv:2507.15035, 2025. (Submitted to *Computer Methods and Programs in Biomedicine*)
3. Zeng Z, Neumann M, Yang Y. Robust Frequency Domain Full-Waveform Inversion via HV-Geometry. *IEEE Transactions on Computational Imaging*, 2025, 11: 1271–1282.

Physics-Informed Knowledge Discovery

1. Zeng Z, Bao C, Hu P, *et al.* Reconstruction of Dynamical Systems Without Time Label. *Communications in Computational Physics*, 2026, 39(3): 941–968.
2. Chen Z, Zeng Z, Hu P, *et al.* Weak Collocation Networks: A deep learning approach to reconstruct stochastic dynamics from aggregate data. *Communications in Nonlinear Science and Numerical Simulation*, 2025: 109360. (Co-first Author)
3. Jiang Y, Zeng Z, Yang W, *et al.* STWCR: Weak collocation regression for revealing hidden stochastic dynamics from single trajectory data. *Applied Mathematics and Computation*, 2026, 521: 129958.
4. Zeng Z, Chen J, Shi Z. BlindNO: A Distributional Neural Operator for Dynamical System Reconstruction from Time-Label-Free Data. *arXiv preprint* arXiv:2511.12316, 2025. (Submitted to *SIAM*)

Journal on Scientific Computing)

5. Lu L, Zeng Z, Jiang Y, *et al.* Weak collocation regression method: Fast reveal hidden stochastic dynamics from high-dimensional aggregate data. *Journal of Computational Physics*, 2024, 502: 112799.
6. Guo L, Lu L, Zeng Z, *et al.* Weak Collocation Regression for Inferring Stochastic Dynamics with Lévy Noise. *Communications in Computational Physics*, 2025, 37(5): 1277–1304.

Generative Models

1. Zeng Z, Gan W, Chen J, *et al.* An Efficient Conditional Score-based Filter for High Dimensional Nonlinear Filtering Problems. *arXiv preprint* arXiv:2509.19816, 2025. (Submitted to *Automatica*)
2. Zeng Z, Chen J. Flow Matching Transport for Quasi-Monte Carlo Integration. *arXiv preprint* arXiv:2601.01072, 2026. (Submitted to *Mathematics and Computers in Simulation*)
3. Gan W, Zeng Z, Chen J, *et al.* Kernel Variational Inference Flow for Nonlinear Filtering Problem. *arXiv preprint* arXiv:2509.18589, 2025. (Submitted to *IEEE TAC*)
4. Zhang G, Li Y, Luo R, *et al.* UniGenX: Unified generation of sequence and structure with autoregressive diffusion. *arXiv preprint* arXiv:2503.06687, 2025.

Language Models

1. Zeng Z, Jiang Y, Hu P, *et al.* Constant-Target Energy Matching: A Unified Framework for Continuous and Discrete Density Estimation. *arXiv preprint* arXiv:2605.09085, 2026. (Submitted to *NeurIPS 2026*)
2. Chen S, Ying J, *et al.* Imaging-101: Benchmarking LLM Agents for Computational Imaging. (Accepted by *ICCP 2026*)

Academic Service

AI4S Cup – Acoustic Field Prediction in Ultrasound CT Imaging (https://www.bohrium.com/competitions/2512153120), Organizer/Reviewer, <i>AI for Science Institute, Beijing (AIS)</i> ; <i>Peking University; Institute of Acoustics, Chinese Academy of Sciences</i>	Mar–Jun 2024
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Teaching Experience

Ordinary Differential Equations , Teaching Assistant, Tsinghua University	Fall 2022
Analytical Mechanics , Teaching Assistant, Tsinghua University	Spring 2023
Advanced Numerical Analysis , Teaching Assistant, Tsinghua University	Fall 2023
Optimization Methods in Machine Learning , Teaching Assistant, Tsinghua University	Spring 2024
Advanced Numerical Analysis , Teaching Assistant, Tsinghua University	Fall 2025

Talks

Neural Born Series Operator for Biomedical Ultrasound CT, International Workshop of Computational Imaging <i>IMSI, Chicago</i>	Aug 2024
In vivo 3D ultrasound computed tomography of musculoskeletal tissues with generative neural physics, CSML 2025 <i>Beijing</i>	Aug 2025
Ultrasound Tomography of Musculoskeletal Tissues with Generative Neural Physics, International Workshop on Medical Ultrasound Tomography <i>HUST, Wuhan</i>	Jun 2026
Ultrasound Tomography of Musculoskeletal Tissues with Generative Neural Physics, ICCP 2026 <i>Princeton</i>	Jul 2026

Posters

In vivo 3D ultrasound computed tomography of musculoskeletal tissues with generative neural physics, ICCP 2025 <i>Toronto</i>	Jul 2025
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